

Hypothesis Testing with E-values - by R. Ramdas and B. Wang

Reading Group 2026

Chapter 9 - FDR control
using compound e-values
Part 4/3

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In the last episode...

- For any $A \subseteq \mathcal{H}$ define $\mathcal{P}_A = \bigcap_{h \in A} \mathcal{P}_h$

The set $\{E_A\}_{A \subseteq \mathcal{H}}$ is called an e -collection if E_A is an e -variable for $\mathcal{P}_A$ for every $A \subseteq \mathcal{H}$.

Example If we have 1 e -value per hypothesis

$$E_A = \frac{1}{|A|} \sum_{h \in A} E_h \quad A \subseteq \mathcal{H}$$

we call those one the mean e -collection

In the last episode

- For any set $A \subseteq \mathcal{H}$ and $R \subseteq \mathcal{H}$, define

$$\text{FDP}_A(R) = \frac{|A \cap R|}{|R| + 1}$$

which is the false discovery proportion of R is the true nulls were A .

In the last episode

Def 9.31 (closed ϵ -BH procedure)

- Let $\{E_A\}_{A \subseteq \mathcal{X}}$ be an ϵ -collection. Define the candidate discovery

$$C_\alpha = \left\{ R \subseteq \mathcal{X} : E_A \geq \frac{FDP_A(R)}{\alpha} \quad \forall A \subseteq \mathcal{X} \right\}$$

- Define an ϵ -closed procedure as a procedure that output some set $R \in C_\alpha$.
- Finally, define the closed ϵ -BH procedure ($\overline{\text{e-BH}}$) at level α as the procedure that rejects a target set in C_α (any of them)

In the last episode

- The closed e-BH procedure controls the FDR at level α . Actually we have the post hoc guarantee

$$\forall P \in \mathcal{H}_1 \quad \mathbb{E}^P \left[\sup_{\alpha \in (0,1)} \sup_{R \in C_\alpha} \frac{\text{FDP}_d(R)}{\alpha} \right] \leq 1$$

- The closed e-BH improves the e-BH (ie it does more discovery)
- Every FDR controlling procedure can be written as the closed e-BH procedure applied to an appropriate e-collection

9.6.3 Recovering or improving any post-hoc FDR procedure.

Th 9.36 Consider an arbitrary post-hoc FDR procedure $\mathcal{D}$. Then, there exists an c -collection $\{E_A\}_{A \subseteq \mathcal{K}}$ for which the corresponding c -BH procedure (at any α) recovers or improves $\mathcal{D}$ (at that α).

Beyond False Discovery Rate
Control

8.5 Family-wise error rate and merging e-values

Suppose that E_k is an e-variable for $\mathcal{P}_k$, $k \in \mathcal{K}$.

For any set $A \subseteq \mathcal{K}$, we can define

$$E_A = F(E_k, k \in A)$$

for F any e-merging function.

$\{E_A\}_{A \subseteq \mathcal{K}}$ forms an e-collection

Example the mean e-collection

$$E_A = \frac{1}{|A|} \sum_{k \in A} E_k$$

($F =$ arithmetic mean $\mathbb{M}$)

We would like that our procedure control
the **FWER** at level $\alpha \in (0, 1)$ ie

$$\forall IP \in \mathcal{H}_1 \quad IP(\mathcal{D} \cap \mathcal{A}' \neq \emptyset) < \alpha$$

ie the probability of rejecting any null hypothesis
is at most α , for every possible distribution IP

In the following we describe any simple
procedure, that control the FWER based
on the e-collection $(E_A)_{A \subseteq \mathcal{K}}$

1) Let
$$E_b^* = \min_{\substack{A \subseteq \mathcal{K} \\ b \in A}} E_A$$

2) Define the procedure $\mathcal{D}$ that reject all indices $b \in \mathcal{K}$ with $E_b^* \geq \frac{1}{\alpha}$

The The procedure described above controls the FWER at level α

Proof

$$\max_{b \in \mathcal{N}} E_b' = \max_{b \in \mathcal{N}} \min_{A: b \in A} E_A$$

$$\leq \max_{b \in \mathcal{N}} E_{\mathcal{N}} = E_{\mathcal{N}}$$

$$\text{and } \mathbb{P}(\mathcal{D} \cap \mathcal{N} \neq \emptyset) = \mathbb{P}\left(\max_{b \in \mathcal{N}} E_b' \geq \frac{1}{\alpha}\right)$$

$$\leq \mathbb{P}\left(E_{\mathcal{N}} \geq \frac{1}{\alpha}\right) \leq \alpha$$



Computation time:

Team e-collection



$$O(K^2)$$

Product ie. merging



$$O(K)$$

(if the e-values are
independent)

9.6.4 A general ϵ -closure principle for arbitrary error metrics

What we have develop is actually much more far-reaching!

- Let $L_A(R)$ represent an arbitrary error metric that quantify the quality of selected set R when the true null set is A
- We would like to design a procedure D whose rejected sets R satisfy



$$\sup_{P \in \mathcal{P}_1} E^P [L_d(R)] \leq \alpha$$

for any α in some range

We can usually assume that L is

- nonnegative
- bounded (and even within $[0, 1]$)

Examples

• FDR control $L_A = \text{FDP}_A$

• B -familywise error rate (B -FWER)

$$L_A = \mathbb{1}_{|A \cap R| \geq B}$$

• per-family error rate (PFER)

$$L_A = |A \cap K| / K$$

• false discovery exceedance (FDX)

$$\Leftrightarrow \underline{\text{FDP}_A(R)} > \gamma \quad \text{for some } \gamma \in (0, 1)$$

Let $\mathcal{L}$ denote all such error metrics

Given an e-collection $\{E_A\}_{A \subseteq \mathcal{X}}$, we can define the candidate set for a particular L by

$$\mathcal{C}_L(L) = \left\{ R \subseteq \mathcal{X} : E_A \geq \frac{L_A(R)}{\alpha} \quad \forall A \subseteq \mathcal{X} \right\}$$

→ Any set $R \in C_\alpha(L)$ satisfy (🌶️)

→ We could choose the target one to reject

Further, if we begin with any procedure D that guarantee (🌶️), it can be recover or improved by applying the ϵ -closure technique on the ϵ -collection define by

$$E_A = L_A(R) / \alpha$$

In fact we would get the stronger guarantee

$$\sup_{P \in \mathcal{P}} \mathbb{E}^P \left[\sup_{L \in \mathcal{L}} \sup_{\alpha \in (0,1)} \sup_{R \in \mathcal{C}_\alpha(L)} \frac{L_{\mathcal{V}}(R)}{\alpha} \right] \leq 1$$

In particular both the error metric L and the level α can be chosen post-hoc

The End

(Finally !)